

Appendix

Inflammation and Acute Traffic-Related Air Pollution Exposures Among a Cohort of Youth with Type 1 Diabetes; Puett, Yanosky, Mittleman, et al., 2019

Table A1. Correlations among air pollution exposures averaged over the week prior to blood draw

	PM2.5 Mass $\mu\text{g}/\text{m}^3$	EC ng/m^3	ADMS Roads A1-3 $\mu\text{g}/\text{m}^3$
ADMS Roads A1-6 $\mu\text{g}/\text{m}^3$	0.17	0.32	0.86
ADMS Roads A1-3 $\mu\text{g}/\text{m}^3$	0.18	0.29	
EC ng/m^3	0.58		

Table A2. Percent differences in IL-6, hs-CRP, and fibrinogen levels with IQR changes in estimated EC, PM_{2.5} and ADMS roads exposures in fully-adjusted models

Air Pollutant Exposures	IL-6 Percent difference (95%CI)	hs-CRP Percent difference (95%CI)	Fibrinogen Percent difference (95%CI)
Lag Day 7 ADMS 1-6	-0.4(-4.5,3.8)	2.0(-3.2,7.5)	-0.2(-0.9,0.5)
Lag Day 5 ADMS 1-6	-1.7(-5.3,2.0)	2.9(-1.9,8.0)	0.03(-0.6,0.7)
Lag Day 1 ADMS 1-6	-1.0(-4.6,2.8)	1.1(-3.4,6.0)	0.3(-0.3,0.9)
Avg Week ADMS 1-6	-1.5(-5.2,2.4)	2.0(-2.8,7.1)	-0.02(-0.7,0.7)
Avg Week ADMS 1-3	-4.4(-8.8,0.2)	2.6(-3.4,8.9)	-0.3(-0.8,0.8)
Lag Day 7 PM _{2.5}	1.8(-2.5,6.3)	0.9(-4.5,6.5)	0.002(-0.7,0.8)
Lag Day 5 PM _{2.5}	1.7(-2.9,6.6)	3.7(-2.1,9.9)	0.2(-0.6,1.0)
Lag Day 1 PM _{2.5}	1.5(-3.0,6.3)	1.7(-4.0,7.6)	0.3(-0.5,1.1)
Avg Week PM _{2.5}	3.7(-1.3,9.0)	4.4(-1.8,11.1)	0.6(-0.3,1.4)
Avg Week EC	8.9(3.2,14.9)	7.7(1.1,14.7)	0.1(-0.8,1.0)

*Model adjusts for site, age, gender, residential census tract percent below poverty, physical activity, sedentary behavior and smoking

Figure A1. DAG Developed To Assist with Selection of Variables for Air Pollution Exposure and Inflammation Models^{1,2}

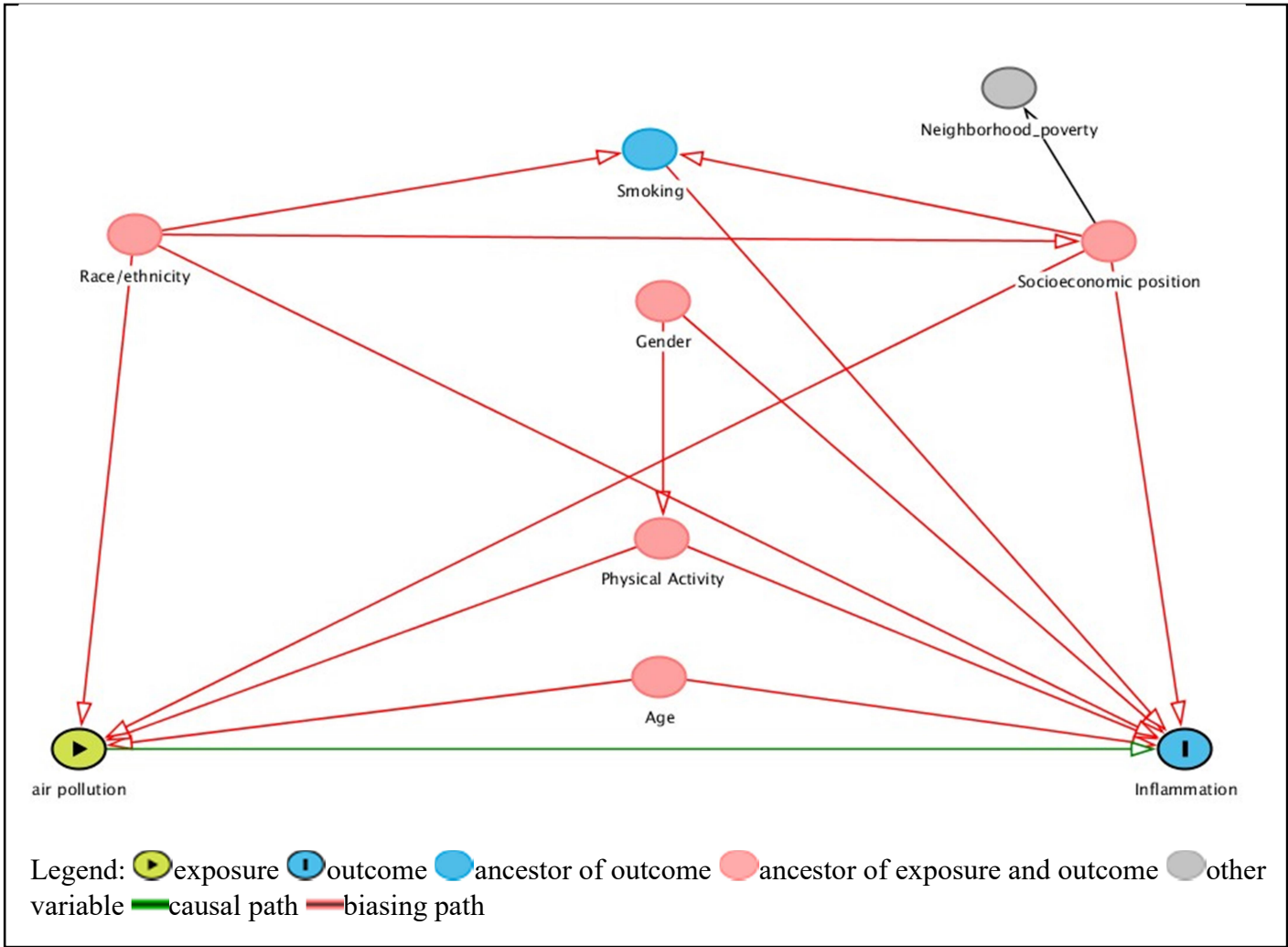


Figure A2. Percent difference in IL-6 levels with an IQR change in estimated in EC, EC, PM_{2.5} and ADMS roads exposures in models adjusted for site only

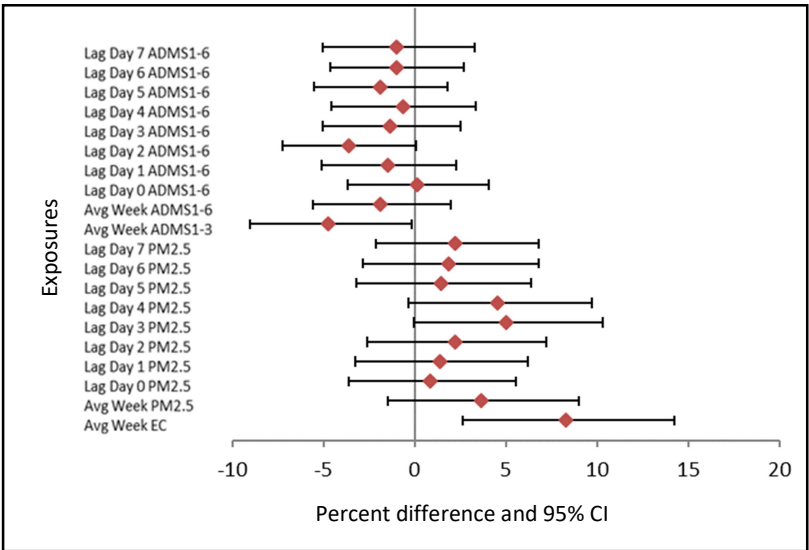
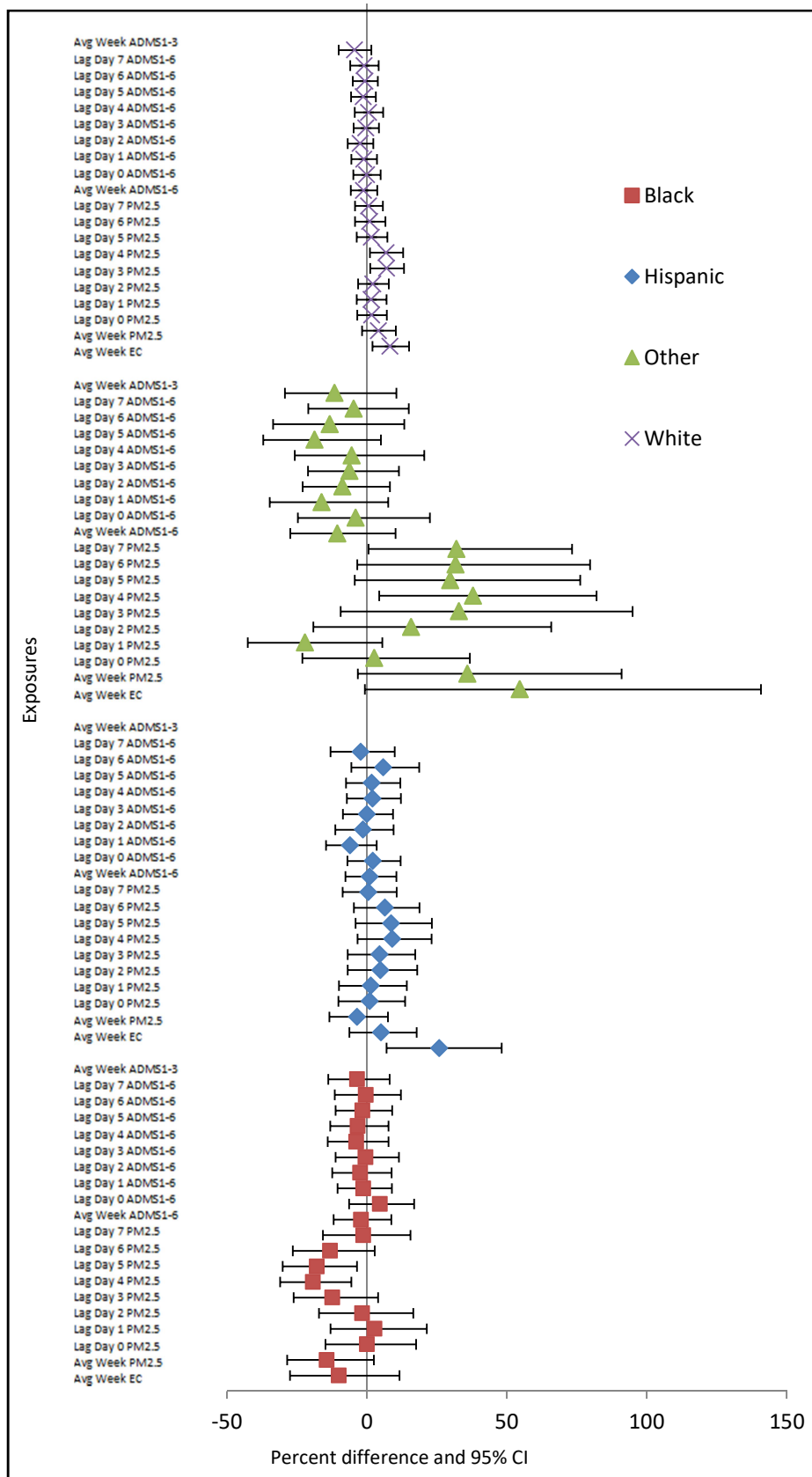


Figure A3. Percent difference in IL6 levels with an IQR change in estimated in EC, PM_{2.5} and ADMS roads exposures in fully-adjusted models stratified by race/ethnicity



*Model adjusts for site, age, gender, residential census tract percent below poverty, physical activity, sedentary behavior and smoking

Figure A4. Percent difference in hs-CRP levels with an IQR change in estimated in EC, PM_{2.5} and ADMS roads exposures in models adjusted for site only

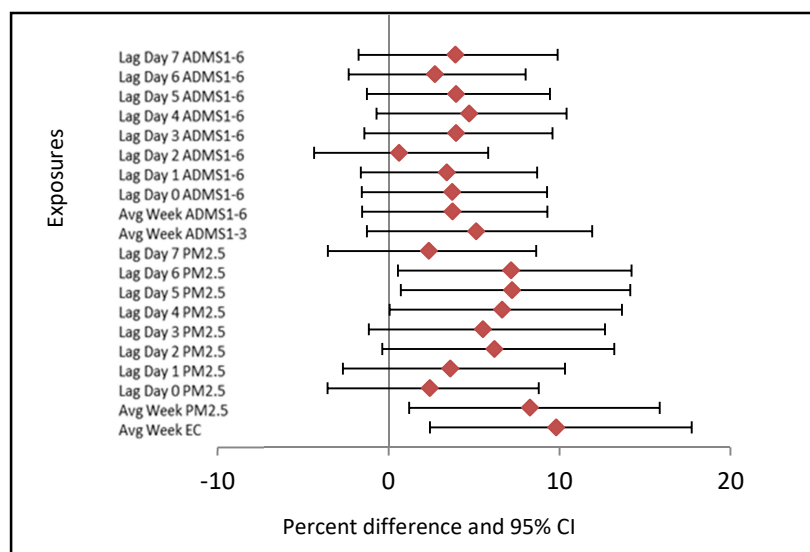
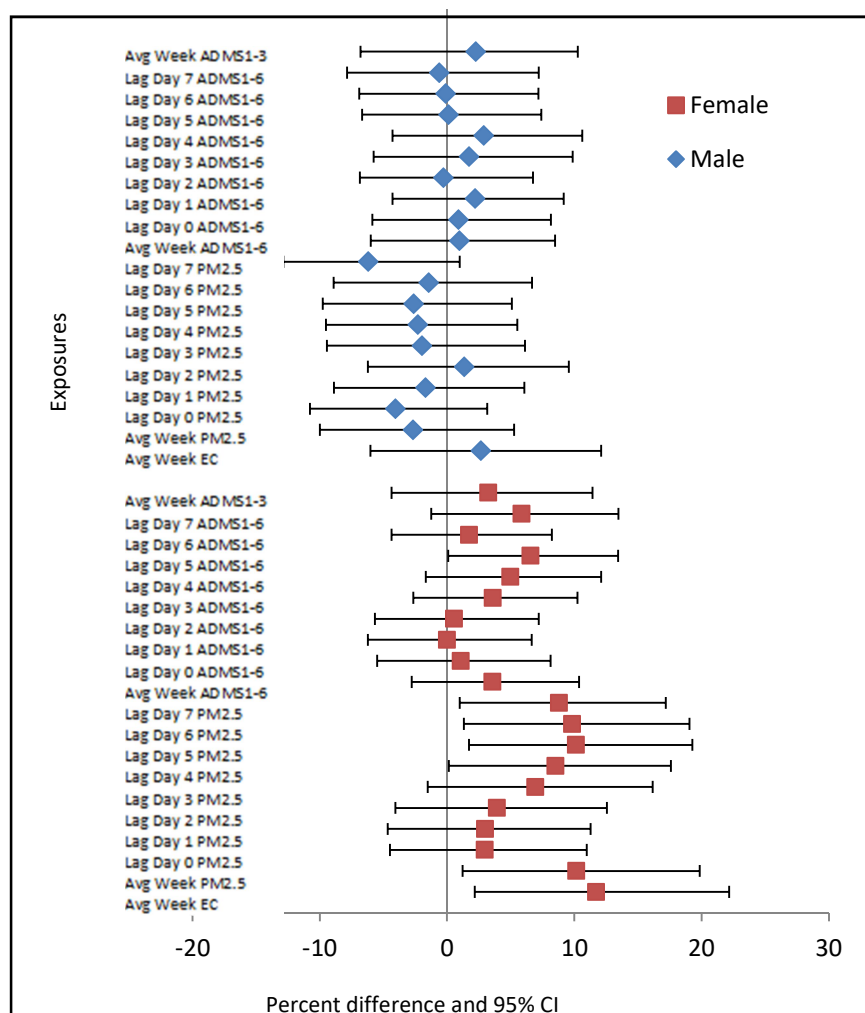


Figure A5 Percent difference in hsCRP levels with an IQR change in estimated in EC, PM_{2.5} and ADMS roads exposures in fully-adjusted models stratified by sex



*Model adjusts for site, age, gender, residential census tract percent below poverty, physical activity, sedentary behavior and smoking.

Figure A6. Percent difference in fibrinogen levels with an IQR change in estimated in EC, PM_{2.5} and ADMS roads exposures in models adjusted for site only

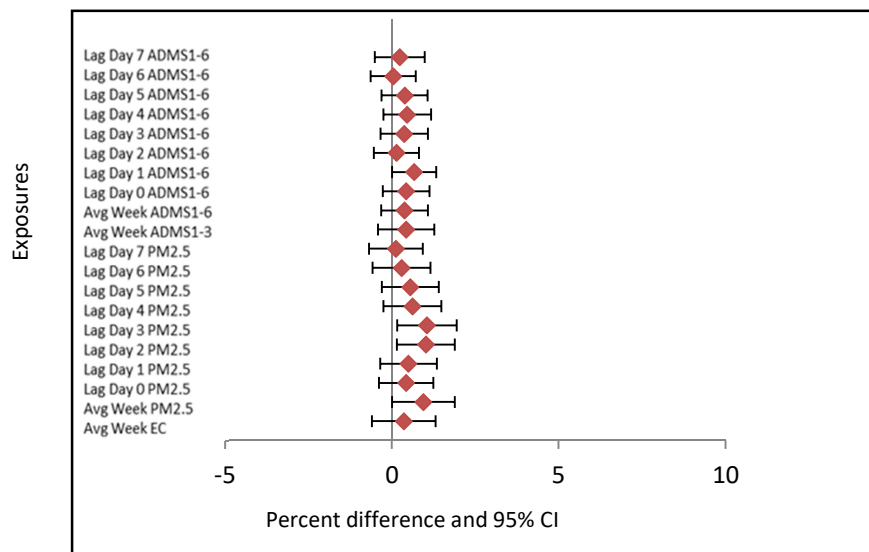
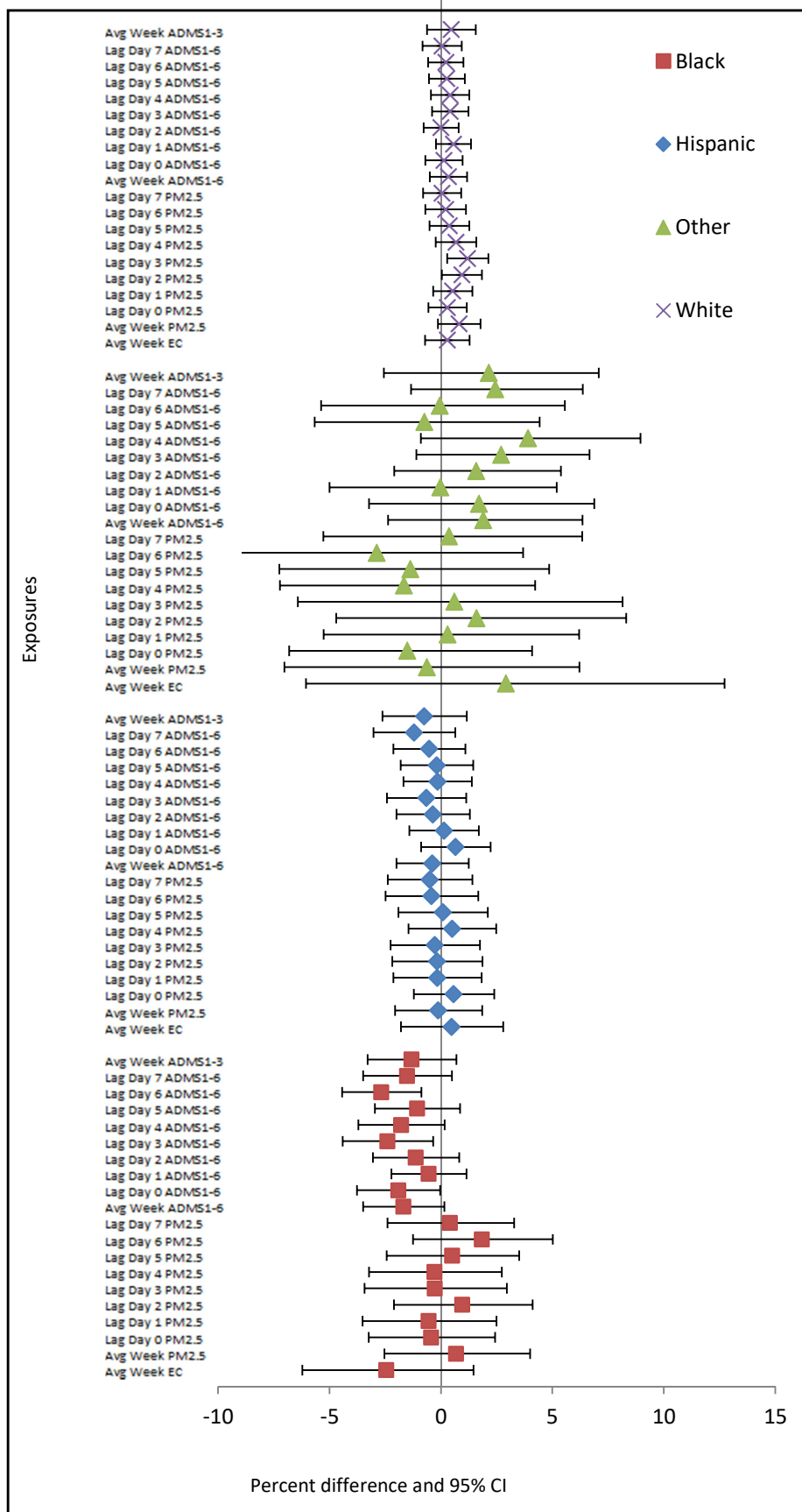


Figure A7 Percent difference in fibrinogen levels with an IQR change in estimated in EC, PM_{2.5} and ADMS roads exposures in fully-adjusted models stratified by race/ethnicity



*Model adjusts for site, age, gender, residential census tract percent below poverty, physical activity, sedentary behavior and smoking

References:

- 1) Daggity.net version 2.3 <http://www.dagitty.net/dags.html#>
- 2) Textor J, Hardt, J, Knuppel S. (2011). Dagitty: A graphical tool for analyzing causal diagrams. *Epidemiology*, 22(5):745.